

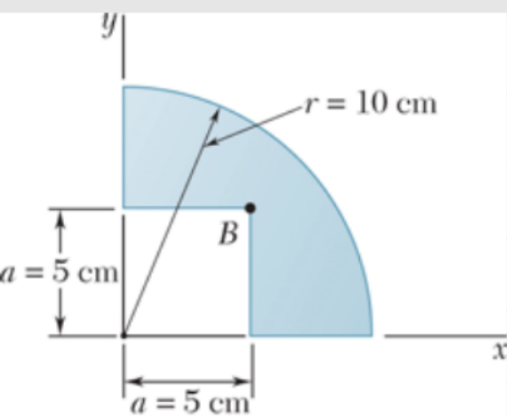
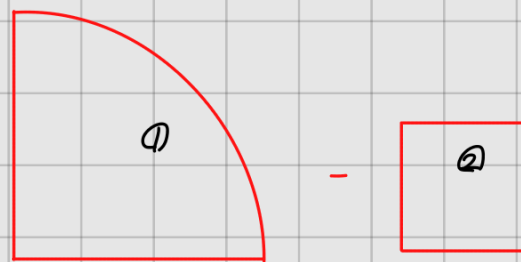


Fig. P5.5



	$\bar{x}$	$\bar{y}$	A	$\bar{x}A$	$\bar{y}A$
①	$\frac{2R}{\pi} = \frac{20}{\pi} \text{ cm}$	$\frac{20}{\pi} \text{ cm}$	$25\pi \text{ cm}^2$	500 cm^3	500 cm^3
②	2.5 cm	2.5 cm	25 cm^2	62.5 cm^3	62.5 cm^3

$$(25\pi - 25) \text{ cm}^2 \quad 437.5 \text{ cm}^3 \quad 437.5 \text{ cm}^3$$

$$\therefore \bar{x}_t = \frac{\sum \bar{x}A}{A} = \frac{437.5}{25\pi - 25} = 8.171 \text{ cm}$$

$$\bar{y}_t = \frac{\sum \bar{y}A}{A} = 8.171 \text{ cm}$$

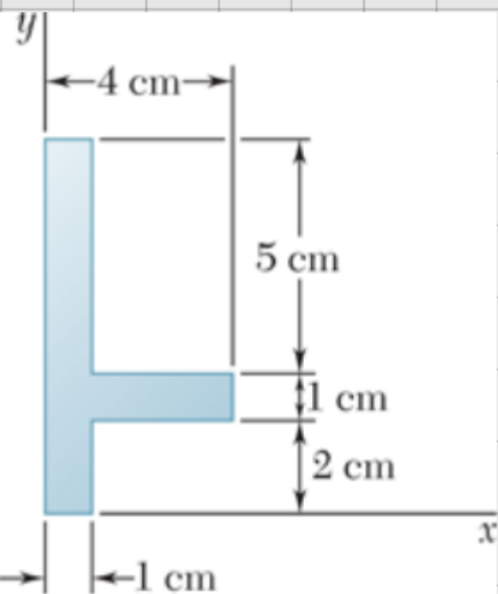
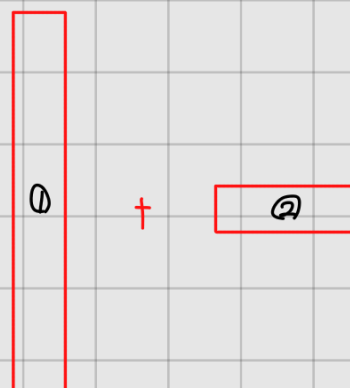


Fig. P5.1



	cm	cm	cm ²	cm ³	cm ³
	$\bar{x}$	$\bar{y}$	A	$\bar{x}A$	$\bar{y}A$
①	0.5	4	8	4	32
②	2.5	2.5	3	7.5	7.5
Σ			11	11.5	39.5

$$\bar{x} = \frac{\sum \bar{x}A}{A} = \frac{11.5}{11} = 1.05 \text{ cm}$$

$$\bar{y} = \frac{\sum \bar{y}A}{A} = \frac{39.5}{11} = 3.59 \text{ cm}$$

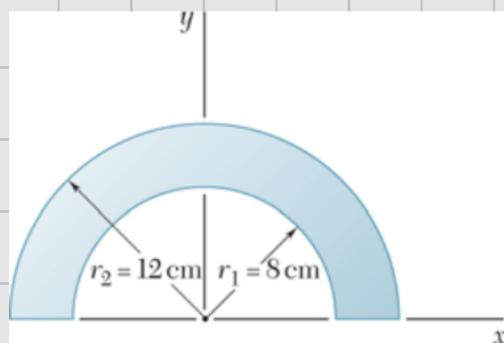
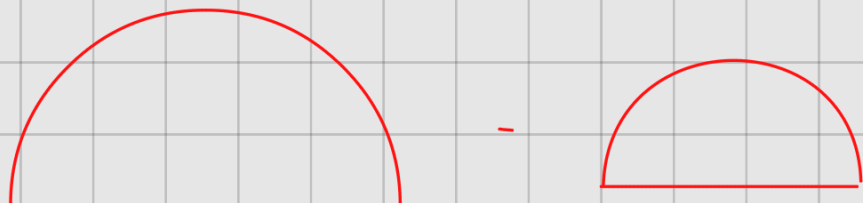


Fig. P5.10



	cm	cm	cm ²	cm ³	cm ³
	$\bar{x}$	$\bar{y}$	A	$\bar{x}A$	$\bar{y}A$
①	0	$\frac{24}{\pi}$	144π	0	3456
②	0	$\frac{16}{\pi}$	64π	0	1024
Σ			80π	0	2432

$$\bar{y} = \frac{\sum \bar{y}A}{A} = \frac{2432}{80\pi} = 9.67 \text{ cm}$$

$$\bar{x} = 0$$

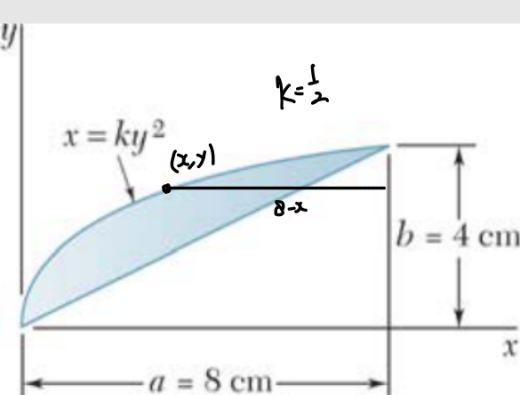
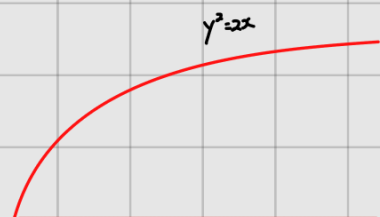


Fig. P5.14



$$2x = y^2$$

$$y = \sqrt{2x}^{1/2}$$

$$\begin{aligned} A_1 &= \int_0^8 y \, dx \\ &= \int_0^8 \sqrt{2x}^{1/2} \, dx \\ &= \frac{2\sqrt{2}}{3} x^{3/2} \Big|_0^8 \\ &= \frac{64}{3} \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \bar{x}_1 A_1 &= \int_0^8 x y \, dx \\ &= \int_0^8 \sqrt{2x}^{3/2} \, dx \\ &= \frac{2\sqrt{2}}{5} x^{5/2} \Big|_0^8 \end{aligned}$$

$$\therefore \bar{x}_1 = 4.8 \text{ cm}$$

$$\begin{aligned} \bar{y}_1 A_1 &= \int_0^4 y(8-x) \, dy \\ &= \int_0^4 8y - xy \, dy \\ &= \int_0^4 8y - \frac{y^3}{2} \, dy \\ &= 4y^2 - \frac{y^4}{8} \Big|_0^4 \end{aligned}$$

$$\bar{y}_1 = 1.5 \text{ cm}$$

$$A_2 = 8 \times 4 \times \frac{1}{2} = 16 \text{ cm}^2$$

$$\bar{x}_2 = 8 \times \frac{2}{3} = \frac{16}{3} \text{ cm}$$

$$\bar{y}_2 = 4 \times \frac{1}{3} = \frac{4}{3} \text{ cm}$$

$$A = A_1 - A_2 = \frac{16}{3} \text{ cm}^2$$

$$\therefore \bar{x} A = \bar{x}_1 A_1 - \bar{x}_2 A_2$$

$$\bar{y} A = \bar{y}_1 A_1 - \bar{y}_2 A_2$$

$$\bar{x} = 3.2 \text{ cm}$$

$$\bar{y} = 2 \text{ cm}$$

Problem 5.28: The homogeneous wire $ABCD$ is bent as shown and is attached to a hinge at C . Determine the length L for which portion BCD of the wire is horizontal.

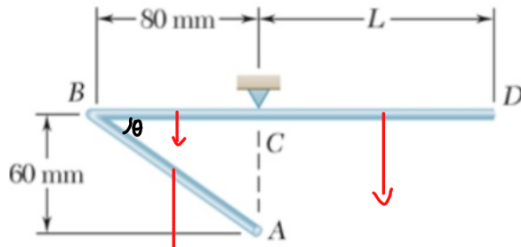


Fig. P5.28 and P5.29

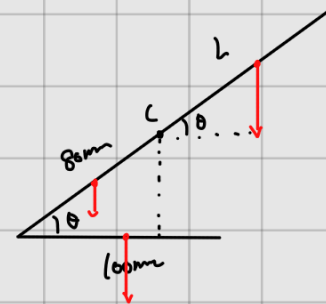
Problem 5.29: The homogeneous wire $ABCD$ is bent as shown and is attached to a hinge at C . Determine the length L for which portion AB of the wire is horizontal.

5.28.

$$\begin{aligned}\sum M_C &= \frac{80 \text{ mm}}{2} \times \rho A (80 \text{ mm}) g + \frac{80 \text{ mm}}{2} \times \rho A (\sqrt{80^2 + 60^2}) g \\ &\quad - \frac{L}{2} \times \rho A L g = 0 \\ 40 \text{ mm} \times 80 \text{ mm} + 40 \text{ mm} \times 100 \text{ mm} - \frac{L^2}{2} &= 0 \\ \therefore L &= 120 \text{ mm}\end{aligned}$$

5.29

$$\theta = \tan^{-1} \left(\frac{60}{80} \right) = 36.864^\circ$$



$$\begin{aligned}\sum M_C &= (80 \cos \theta - 50) \times \rho A (100 \text{ mm}) g + 40 \cos \theta \times \rho A (80 \text{ mm}) g - \frac{L}{2} \cos \theta \times \rho A L g = 0 \\ (80 \cos \theta - 50)(100 \text{ mm}) + 40 \cos \theta (80 \text{ mm}) - \frac{L^2}{2} \cos \theta &= 0 \\ \therefore L &= 99.5 \text{ mm}\end{aligned}$$

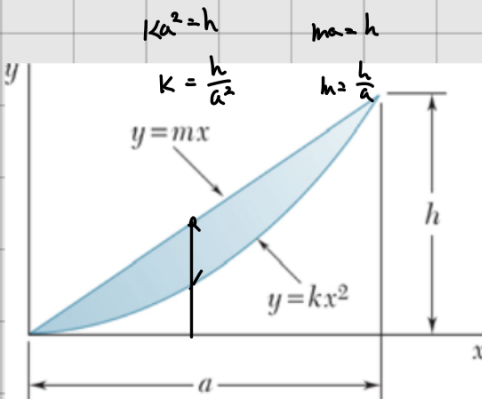


Fig. P5.36

5.36

$$\begin{aligned}A &= \int_0^a y dx \\ &= \int_0^a mx - kx^2 dx \\ &= \left[\frac{1}{2}mx^2 - \frac{1}{3}kx^3 \right]_0^a \\ &= \frac{ma^2}{2} - \frac{ka^3}{3} = \frac{ah}{2} - \frac{ah}{3} = \frac{ah}{6}\end{aligned}$$

$$\begin{aligned}\bar{x} A &= \int_0^a x dA \\ &= \int_0^a x y dx \\ &= \int_0^a x (mx - kx^2) dx \\ &= \left[\frac{mx^3}{3} - \frac{kx^4}{4} \right]_0^a \\ &= \frac{ma^3}{3} - \frac{ka^4}{4} = \frac{a^3h}{3} - \frac{a^4h}{4} = \frac{a^4h}{12}\end{aligned}$$

$$\bar{x} = \frac{a}{2}$$

$$\begin{aligned}\bar{y} A &= \int_0^a \frac{1}{2} (mx + kx^2) dA \\ &= \int_0^a \frac{1}{2} (mx + kx^2) (mx - kx^2) dx \\ &= \int_0^a \left(\frac{m^2x^2}{2} - \frac{k^2x^4}{2} \right) dx \\ &= \left[\frac{m^2x^3}{6} - \frac{k^2x^5}{10} \right]_0^a\end{aligned}$$

$$\begin{aligned}&= \frac{m^2a^3}{6} - \frac{k^2a^5}{10} \\ &= \frac{ah^2}{6} - \frac{ah^2}{10} \\ &= \frac{ah^2}{15} \\ \therefore \bar{y} &= \frac{2h}{5}\end{aligned}$$

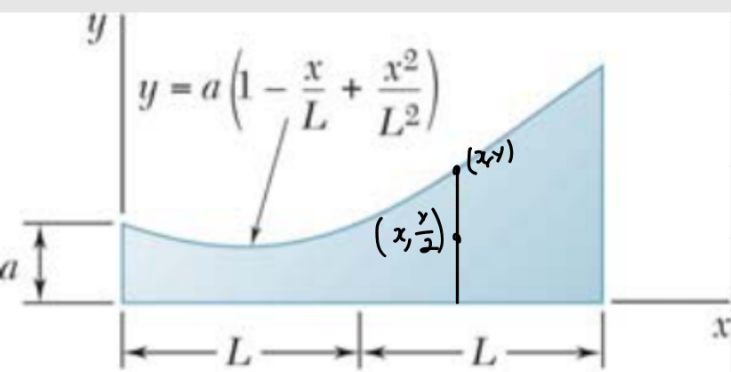


Fig. P5.38

$$\begin{aligned}
 A &= \int_0^{2L} y \, dx \\
 &= \int_0^{2L} a \left(1 - \frac{x}{L} + \frac{x^2}{L^2} \right) dx \\
 &= a \left[x - \frac{1}{2L} x^2 + \frac{1}{3L^2} x^3 \right]_0^{2L} \\
 &= 2aL - 2aL + \frac{8aL}{3} \\
 &= \frac{8aL}{3}
 \end{aligned}$$

$$\begin{aligned}
 \bar{x} A &= \int_0^{2L} x_{\text{ell}} \, dA \\
 &= \int_0^{2L} a x \left(1 - \frac{x}{L} + \frac{x^2}{L^2} \right) dx \\
 &= a \left[\frac{x^2}{2} - \frac{1}{3L} x^3 + \frac{1}{4L^2} x^4 \right]_0^{2L} \\
 &= 2aL^2 - \frac{8}{3} aL^2 + 4aL^2 \\
 &= \frac{10}{3} aL^2
 \end{aligned}$$

$$\bar{x} = \frac{5}{4} L$$

$$\begin{aligned}
 \bar{y} A &= \int_0^{2L} y_{\text{ell}} \, dA \\
 &= \int_0^{2L} \frac{a^2}{2} \left(1 - \frac{x}{L} + \frac{x^2}{L^2} \right)^2 dx \\
 &= \frac{a^2}{2} \int_0^{2L} \left(1 - \frac{2x}{L} + \frac{3x^2}{L^2} - \frac{2x^3}{L^2} + \frac{1}{L^2} x^4 \right) dx \\
 &= \frac{a^2}{2} \left(x - \frac{1}{L} x^2 + \frac{1}{L^2} x^3 - \frac{1}{2L^2} x^4 + \frac{1}{5L^2} x^5 \right) \Big|_0^{2L} \\
 &= \frac{11}{5} a^2 L
 \end{aligned}$$

$$\therefore \bar{y} = \frac{33}{40} a$$

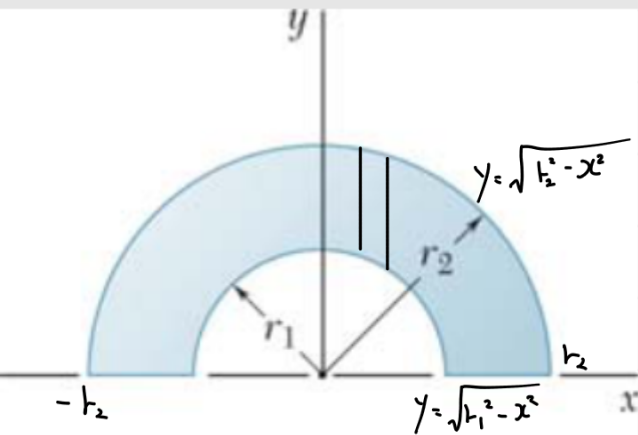


Fig. P5.39

$$A = \frac{\pi r_2^2}{2} - \frac{\pi r_1^2}{2}$$

$$= \frac{\pi}{2} (r_2^2 - r_1^2)$$

중심대칭이므로

$$\bar{x} = 0$$

$$\bar{y} A = \int y_{el} dA$$

$$= \int_0^{r_2} (\sqrt{r_2^2 - x^2} + \sqrt{r_1^2 - x^2}) (\sqrt{r_2^2 - x^2} - \sqrt{r_1^2 - x^2}) dx$$

$$= \int_0^{r_2} (r_2^2 - x^2 - r_1^2 + x^2) dx$$

$$= (r_2^2 - r_1^2)x \Big|_0^{r_2}$$

$$= (r_2^2 - r_1^2)r_2$$

$$\therefore \bar{y} = \frac{2}{\pi} r_2$$

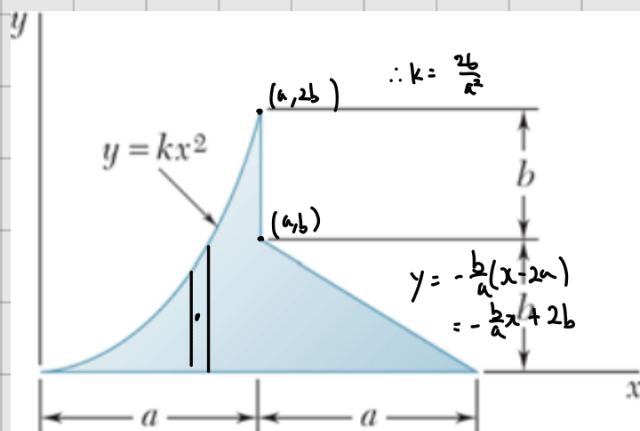


Fig. P5.43

$$A = \int_0^a kx^2 dx + \int_a^{2a} \left(-\frac{b}{a}x + 2b\right) dx$$

$$= \frac{kx^3}{3} \Big|_0^a + \left(-\frac{b}{2a}x^2 + 2bx\right) \Big|_a^{2a}$$

$$= \frac{k}{3}a^3 + (-2ab + 4ab) - \left(-\frac{ab}{2} + 2ab\right)$$

$$= \frac{k}{3}a^3 + \frac{1}{2}ab$$

$$= \frac{2ab}{3} + \frac{1}{2}ab = \frac{7}{6}ab$$

$$\bar{y} A = \int y_{el} dA$$

$$= \int_0^a kx^3 dx + \int_a^{2a} \left(-\frac{b}{a}x^2 + 2bx\right) dx$$

$$= \frac{1}{4}kx^4 \Big|_0^a + \left(-\frac{b}{3a}x^3 + bx^2\right) \Big|_a^{2a}$$

$$= \frac{ka^4}{4} + \left(-\frac{8a^3b}{3a} + 4a^2b\right) - \left(-\frac{a^3b}{3a} + a^2b\right)$$

$$= \frac{a^2b}{2} + \frac{4}{3}a^2b - \frac{2}{3}a^2b$$

$$= \frac{7}{6}a^2b$$

$$\bar{y} = a$$

$$\bar{y} A = \int y_{el} dA$$

$$= \int_0^a \frac{kx^4}{2} dx + \int_a^{2a} \frac{1}{2} \left(-\frac{b}{a}x + 2b\right)^2 dx$$

$$= \frac{k^2}{10}x^5 \Big|_0^a + \frac{b^2}{6a^2}x^3 - \frac{4b^2}{a}x^2 + 2b^2x \Big|_a^{2a}$$

$$= \frac{k^2a^5}{10} + \left(\frac{4}{3}ab^2 - 4ab^2 + 4ab^2\right) - \left(\frac{ab^2}{6} - ab^2 + 2ab^2\right)$$

$$= \frac{2a^2b^2}{5} + \frac{4}{3}ab^2 - \frac{7}{6}ab^2$$

$$= \frac{17}{30}a^2b^2$$

$$\therefore \bar{y} = \frac{17}{35}b$$

Problem 5.45: A homogeneous wire is bent into the shape as shown. Determine by direct integration the x coordinate of its centroid.

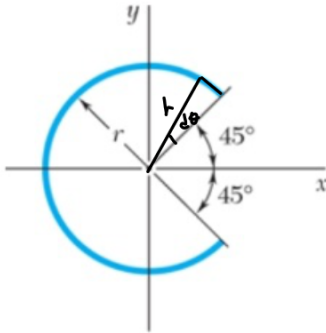


Fig. P5.45

x 좌표 찾기

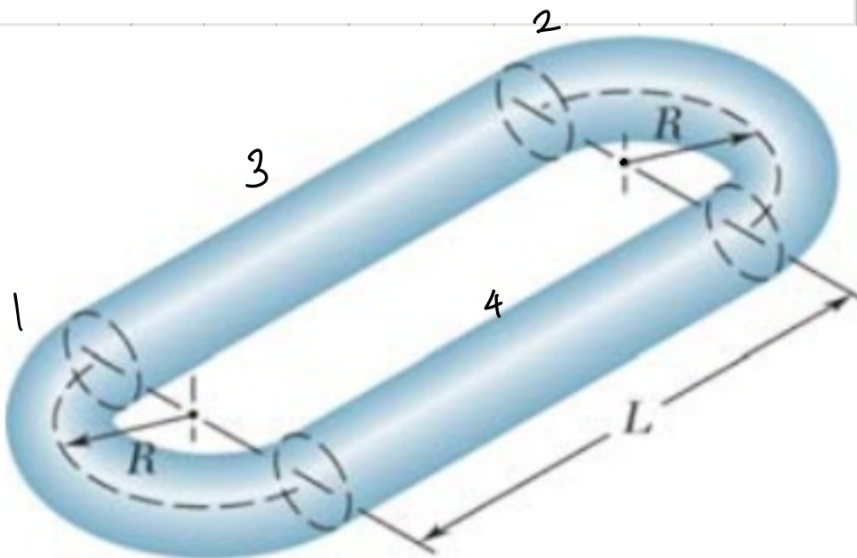
$$\bar{y} = 0$$

$$dL = r d\theta$$

$$\begin{aligned} L &= \int dL = \int_{-\pi/4}^{\pi/4} r d\theta \\ &= r \theta \Big|_{-\pi/4}^{\pi/4} \\ &= \frac{3\pi r}{2} \end{aligned}$$

$$\begin{aligned} \bar{x} L &= \int x dL \\ &= \int_{-\pi/4}^{\pi/4} x r d\theta \\ &= \int_{-\pi/4}^{\pi/4} r^2 \cos\theta d\theta \\ &= r^2 \sin\theta \Big|_{-\pi/4}^{\pi/4} \\ &= -\sqrt{2} r^2 \\ \bar{x} &= \frac{-2\sqrt{2}}{3\pi} L \end{aligned}$$

Problem 5.56: Determine the volume and the surface area of the chain link as shown, which is made from a 6-mm-diameter bar, if $R = 10$ mm and $L = 30$ mm.



$$A_1 = A_3 = \int_0^\pi \int dA d\theta = \int_0^\pi \int r dL d\theta = \pi R(RL) = 60\pi^2$$

$$A_2 = A_4 = AL = \pi dL = 180\pi$$

$$\therefore A = A_1 + A_2 + A_3 + A_4 = 2315 \text{ mm}^2$$

$$\begin{aligned} V_1 = V_3 &= \int_0^\pi \int dV d\theta = \int_0^\pi \int r dA d\theta \\ &= \int_0^\pi R A d\theta = 2R \times \frac{\pi}{4} L^2 \\ &= \frac{\pi^2}{4} R L^2 \end{aligned}$$

$$V_2 = V_4 = AL = \frac{\pi d^2 L}{4}$$

$$V = V_1 + V_2 + V_3 + V_4 = 3413 \text{ mm}^3$$

Problem 5.58, 5.59: Determine the volume and weight of the solid brass knob knowing that the density of the brass is 8470 kg/m^3 and determine the total surface area of the solid brass knob.

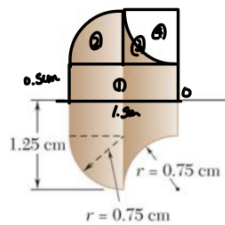


Fig. P5.58 and P5.59

$$① + ② + ③ - ④$$

$$①: \bar{x}_1 = -0.75 \text{ cm} \quad A_1 = 0.75 \text{ cm}^2$$

$$\bar{y}_1 = 0.25 \text{ cm}$$

$$②: \bar{x}_2 = -0.75 - \frac{4 \times 0.75}{3\pi} = -1.068 \text{ cm}$$

$$A_2 = 0.4417 \text{ cm}^2$$

$$\bar{y}_2 = 0.5 + \frac{4 \times 0.75}{3\pi} = 0.8183 \text{ cm}$$

$$③: \bar{x}_3 = -0.75 + \frac{0.75}{2} = -0.375 \text{ cm}$$

$$A_3 = 0.5625 \text{ cm}^2$$

$$\bar{y}_3 = 0.5 + \frac{0.75}{2} = 0.875 \text{ cm}$$

$$④: \bar{x}_4 = -\frac{4 \times 0.75}{3\pi} = -0.3183 \text{ cm}$$

$$A_4 = 0.4417 \text{ cm}^2$$

$$\bar{y}_4 = 1.25 - \frac{4 \times 0.75}{3\pi} = 0.9316 \text{ cm}$$

$$A = A_1 + A_2 + A_3 - A_4 = 1.3125 \text{ cm}^2$$

$$\bar{x}A = \sum \bar{x}_i A_i$$

$$\bar{x} = -1.055 \text{ cm}$$

$$\bar{y}A = \sum \bar{y}_i A_i$$

$$\bar{y} = 0.4797 \text{ cm}$$

$$\therefore V = \int_0^{2\pi} \int_0^{\pi} r^2 \sin \theta \, d\theta \, d\phi = 2\pi \bar{y}A = 3.956 \text{ cm}^3$$

$$W = \rho V g = 8470 \times 3.956 \times 10^{-6} \times 9.81 = 0.328 \text{ N}$$

Problem 5.58, 5.59: Determine the volume and weight of the solid brass knob knowing that the density of the brass is 8470 kg/m^3 and determine the total surface area of the solid brass knob.

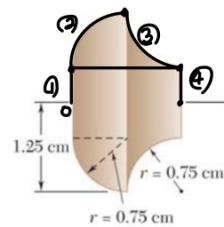


Fig. P5.58 and P5.59

$$① + ② + ③ + ④$$

$$①: \bar{y}_1 = 0.25 \text{ cm}$$

$$L_1 = 0.5 \text{ cm}$$

$$②: \bar{y}_2 = 0.5 + \frac{2 \times 0.75}{\pi} = 0.9174 \text{ cm}$$

$$L_2 = 1.178 \text{ cm}$$

$$③: \bar{y}_3 = 1.25 - \frac{2 \times 0.75}{\pi} = 0.9125 \text{ cm}$$

$$L_3 = 1.178 \text{ cm}$$

$$④: \bar{y}_4 = 0.25 \text{ cm}$$

$$L_4 = 0.5 \text{ cm}$$

$$L = L_1 + L_2 + L_3 + L_4 = 3.386 \text{ cm}$$

$$\bar{y}L = \sum \bar{y}_i L_i$$

$$\bar{y} = 0.6887 \text{ cm}$$

$$\therefore A = \int_0^{2\pi} \int_0^{\pi} r \sin \theta \, d\theta \, d\phi = \int_0^{2\pi} \bar{y}L \, d\phi = 2\pi \bar{y}L$$

$$= 14.52 \text{ cm}^2$$

Problem 5.68, 5.70, 5.72: Determine the reactions at the beam supports for the given loading. $\frac{2600}{3}x = y$

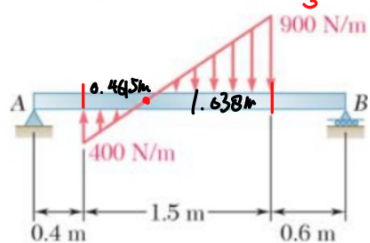


Fig. P5.68

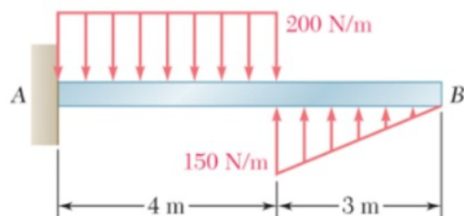


Fig. P5.70

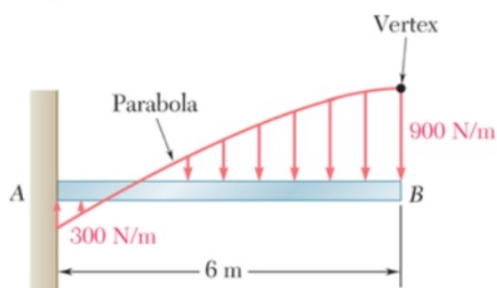
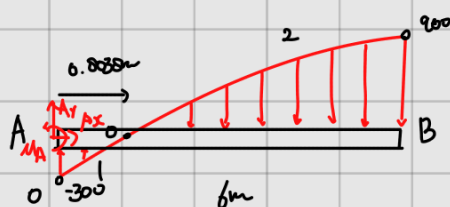


Fig. P5.72



$$y = a(x-6)^2 + 900$$

$$300 = 36a + 900$$

$$a = -\frac{100}{3}$$

$$y = -\frac{100}{3}(x-6)^2 + 900$$

$$y = 0 \text{ at } x = 0.8038$$

$$\therefore \int_0^{0.8038} -\frac{100}{3}(x-6)^2 + 900 \, dx$$

$$= 117.6 \text{ N} \uparrow$$

$$\bar{x}_1 A_1 = \int x \, dA$$

$$\bar{x}_1 A_1 = \int_0^{0.8038} x \left(-\frac{100}{3}(x-6)^2 + 900 \right) dx$$

$$\bar{x}_1 = 0.2648 \text{ m}$$

$$2: \int_{0.8038}^6 -\frac{100}{3}(x-6)^2 + 900 \, dx$$

$$= 3117.6 \text{ N} \downarrow$$

$$\bar{x}_2 A_2 = \int x \, dA$$

$$= \int_{0.8038}^6 x \left(-\frac{100}{3}(x-6)^2 + 900 \right) dx$$

$$\bar{x}_2 = 4.651 \text{ m}$$

$$\sum F_x = A_2 = 0$$

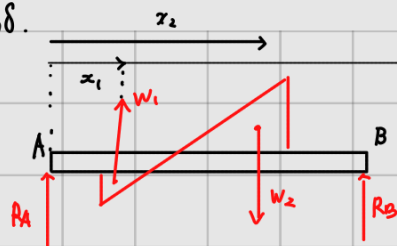
$$\sum F_y = A_y + F_1 - F_2 = 0$$

$$\sum M_A = M_A + F_1 \bar{x}_1 - F_2 \bar{x}_2 = 0$$

$$\therefore A_y = 3000 \text{ N} \uparrow$$

$$M_A = 12598 \text{ N}\cdot\text{m} \curvearrowright$$

5.68.



$$W_1 = 400 \times 0.4615 \times \frac{1}{2} = 92.3 \text{ N}$$

$$W_2 = 900 \times 1.038 \times \frac{1}{2} = 467.1 \text{ N}$$

$$x_1 = 0.4 + 0.4615 \times \frac{1}{3} = 0.5538 \text{ m}$$

$$x_2 = 0.4 + 0.4615 + 1.038 \times \frac{2}{3} = 1.5535 \text{ m}$$

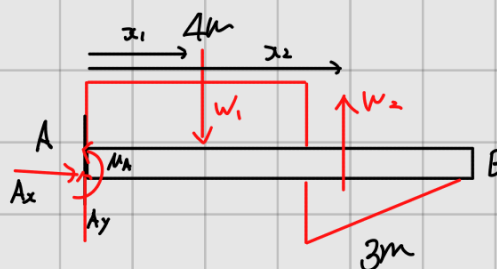
$$\sum F_y = R_A + R_B + W_1 - W_2 = 0$$

$$R_A + R_B = 376.8 \text{ N}$$

$$\sum M_A = W_1 x_1 - W_2 x_2 + R_B \times 2.5 \text{ m} = 0$$

$$\therefore R_B = 264.8 \text{ N}$$

$$\therefore R_A = 111.0 \text{ N}$$



$$W_1 = 200 \times 4 = 800 \text{ N}$$

$$x_1 = 2 \text{ m}$$

$$W_2 = 150 \times 3 \times \frac{1}{2} = 225 \text{ N}$$

$$x_2 = 4 + 3 \times \frac{1}{3} = 5 \text{ m}$$

$$\sum F_x = A_x = 0$$

$$\sum F_y = A_y - W_1 + W_2 = 0$$

$$\sum M_A = M_A - W_1 x_1 + W_2 x_2 = 0$$

$$\therefore A_y = 575 \text{ N}$$

$$M_A = 475 \text{ N}\cdot\text{m}$$

$$A_x = 0 \text{ N}$$

Problem 5.74: Determine the reactions at the beam supports for the given loading.

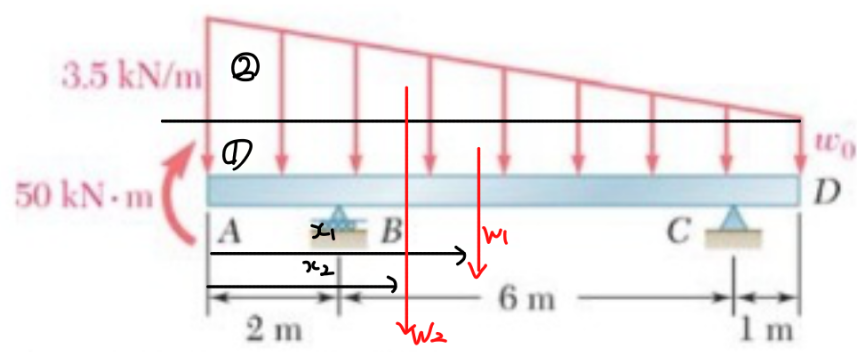


Fig. P5.74 and P5.75

Problem 5.100, 5.103: For the machine element shown, locate the y and z coordinates of the center of gravity.

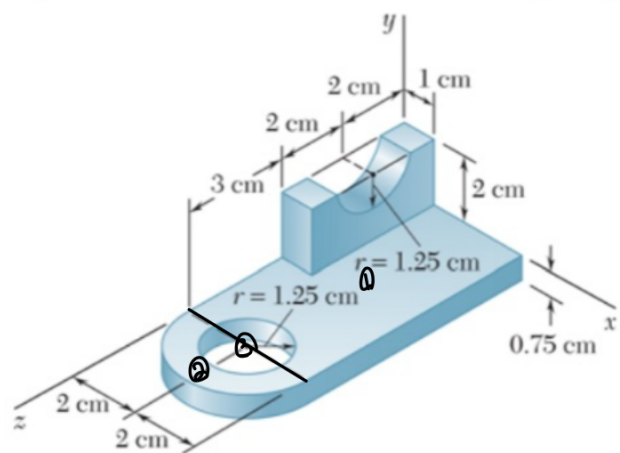


Fig. P5.100 and P5.103

$$\textcircled{1} : W_1 = 9w_0 \quad x_1 = \frac{9}{2} = 4.5 \text{ m}$$

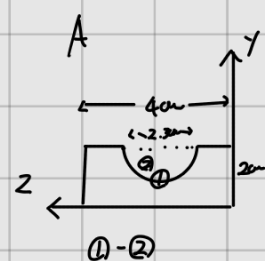
$$\textcircled{2} : W_2 = 4.5(3.5 - w_0) \quad x_2 = \frac{1}{3} \times 9 = 3 \text{ m}$$

$$\Sigma F_y = B_y + C_y - W_1 - W_2 = 0$$

$$\Sigma M_B = -(x_1 - 2) \times W_1 - (x_2 - 2)W_2 + 6C_y = 0$$

$$\therefore C_y = 3w_0 + 2.625 \text{ kN}$$

$$B_y = w_0 + 13.125 \text{ kN}$$



$$\textcircled{1} : \bar{y}_1 = 1 \text{ cm} \quad V_1 = 8 \text{ cm}^2$$

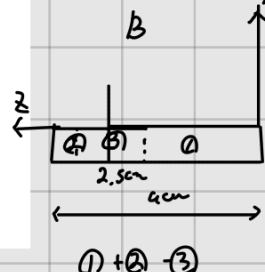
$$\bar{z}_1 = 2 \text{ cm}$$

$$\textcircled{2} : \bar{y}_2 = 2 - \frac{4 \times 1.25}{3\pi} = 1.464 \text{ cm}$$

$$z_2 = 2 \text{ cm} \quad V_2 = 1.963 \text{ cm}^2$$

$$\therefore \bar{y}_A V_A = 5.114 \quad V_A = V_1 - V_2 = 6.037 \text{ cm}^2$$

$$\bar{z}_A V_A = 12.074$$



$$\textcircled{1} : \bar{y}_1 = -0.75/2 = -0.375 \text{ cm}$$

$$\bar{z}_1 = 7/2 = 3.5 \text{ cm}$$

$$V_1 = \pi \times 4 \times 0.75 = 21 \text{ cm}^2$$

$$\textcircled{2} : \bar{y}_2 = -0.75/2 = -0.375 \text{ cm}$$

$$z_2 = 7 + \frac{4 \times 2}{3\pi} = 7.848 \text{ cm}$$

$$V_2 = 4\pi \times \frac{1}{2} \times 0.75 = 4.712 \text{ cm}^2$$

$$\textcircled{3} : \bar{y}_3 = -0.75/2 = -0.375 \text{ cm}$$

$$\bar{z}_3 = 7 \text{ cm}$$

$$V_3 = 1.25^2 \pi \times 0.75 = 3.681 \text{ cm}^2$$

$$V_B = V_1 + V_2 - V_3 = 22.031$$

$$\bar{y}_B V_B = -8.261$$

$$\bar{z}_B V_B = 84.712$$

$$\therefore \bar{y} = \frac{\bar{y}_B V_B + \bar{y}_A V_A}{V} \quad V = V_A + V_B = 28.068 \text{ cm}^2$$

$$= -0.1121 \text{ cm}$$

$$\bar{z} = \frac{\bar{z}_B V_B + \bar{z}_A V_A}{V}$$

$$= 3.448 \text{ cm}$$

Problem 5.107: Locate the center of gravity of the sheet-metal form shown.

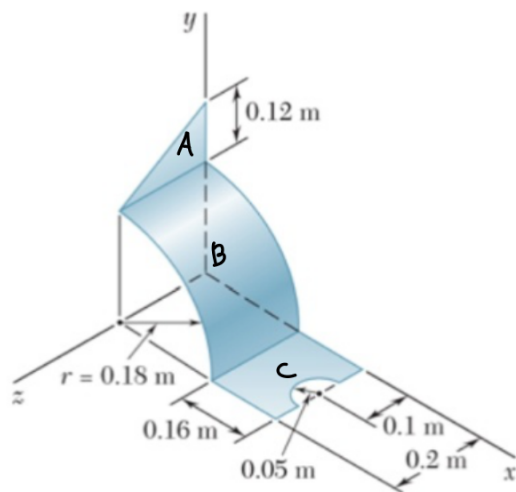
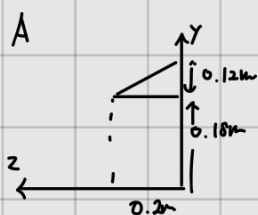


Fig. P5.107

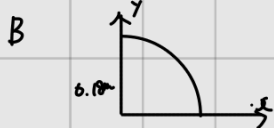


$$\bar{x}_A = 0 \text{ m}$$

$$\bar{y}_A = 0.18 + 0.12 \times \frac{1}{3} = 0.22 \text{ m}$$

$$\bar{z}_A = 0.2 \times \frac{1}{3} = \frac{1}{15} \text{ m}$$

$$A_A = \frac{3}{280} \text{ m}^2$$

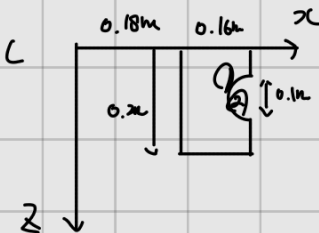


$$\bar{x}_B = \frac{4 \times 0.18}{3\pi} = 0.67639 \text{ m}$$

$$\bar{y}_B = \frac{4 \times 0.18}{3\pi} = 0.67639 \text{ m}$$

$$\bar{z}_B = 0.1 \text{ m}$$

$$A_B = 2\pi \times 0.18 \times \frac{1}{4} \times 0.2 = 0.6565 \text{ m}^2$$



① - ②

$$\bar{x}_1 = 0.18 + 0.16 \times \frac{1}{2} = 0.26 \text{ m}$$

$$\bar{z}_1 = 0.2/2 = 0.1 \text{ m}$$

$$A_1 = 0.032 \text{ m}^2$$

$$\bar{x}_2 = 0.18 + 0.16 - \frac{4 \times 0.05}{3\pi}$$

$$= 0.3187 \text{ m}$$

$$\bar{z}_2 = 0.1 \text{ m}$$

$$A_2 = 3.926 \times 10^{-3} \text{ m}^2$$

$$A_C = A_1 - A_2 = 0.02867 \text{ m}^2$$

$$\begin{aligned} \bar{x}_C A_C &= \bar{x}_1 A_1 - \bar{x}_2 A_2 \\ &= 7.668 \times 10^{-3} \end{aligned}$$

$$\begin{aligned} \bar{z}_C A_C &= \bar{z}_1 A_1 - \bar{z}_2 A_2 \\ &= 0.03147 \end{aligned}$$

$$\begin{aligned} A &= A_A + A_B + A_C \\ &= 0.69661 \end{aligned}$$

$$\bar{x} = \frac{\sum \bar{x}_i A_i}{A}$$

$$= 0.1178 \text{ m}$$

$$\bar{y} = \frac{\sum \bar{y}_i A_i}{A}$$

$$= 0.07203 \text{ m}$$

$$\bar{z} = \frac{\sum \bar{z}_i A_i}{A}$$

$$= 0.3925 \text{ m}$$

Problem 5.115: Locate the center of gravity of the figure shown, knowing that it is made of thin brass rods of uniform diameter.

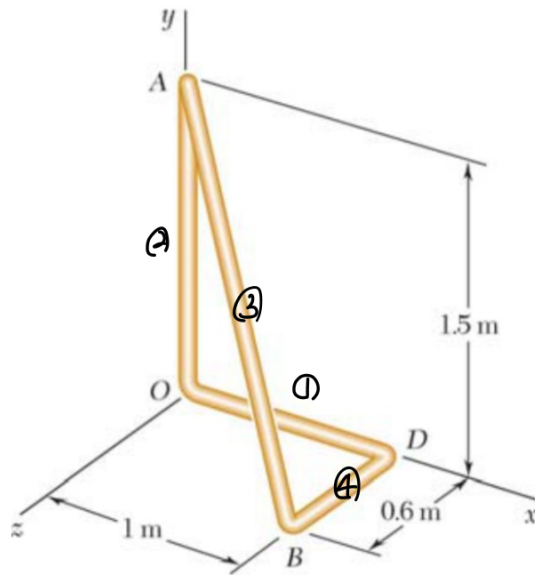


Fig. P5.115

$$\textcircled{1} \quad \bar{x}_1 = 0.5 \text{ m}$$

$$\bar{y}_1 = 0$$

$$\bar{z}_1 = 0$$

$$L_1 = 1.5 \text{ m}$$

$$\textcircled{2} \quad \bar{x}_2 = 0$$

$$\bar{y}_2 = 1.5/2 = 0.75 \text{ m}$$

$$\bar{z}_2 = 0$$

$$L_2 = 1.5 \text{ m}$$

$$\textcircled{3} \quad \bar{x}_3 = 0.5 \text{ m}$$

$$\bar{y}_3 = 0.75 \text{ m}$$

$$\bar{z}_3 = 0.3 \text{ m}$$

$$L_3 = \sqrt{1^2 + 0.6^2 + 1.5^2} = 1.9 \text{ m}$$

$$\textcircled{4} \quad \bar{x}_4 = 1 \text{ m}$$

$$\bar{y}_4 = 0 \text{ m}$$

$$\bar{z}_4 = 0.3 \text{ m}$$

$$L_4 = 0.6 \text{ m}$$

$$L = L_1 + L_2 + L_3 + L_4$$

$$= 5.35 \text{ m}$$

$$\bar{x} = \frac{\sum \bar{x}_i L_i}{L}$$

$$= 0.383 \text{ m}$$

$$\bar{y} = \frac{\sum \bar{y}_i L_i}{L}$$

$$= 0.476 \text{ m}$$

$$\bar{z} = \frac{\sum \bar{z}_i L_i}{L}$$

$$= 0.140 \text{ m}$$